Aly Hegazy

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EDUCATION

CUNY College of Staten Island Staten Island, NY

B.S. in Computer Science and Mathematics; GPA: 3.7/4.0 *Expected May 2027*

Relevant Coursework: Data Structures, Data Science, Practical Machine Learning, Probability and Statistics, Computational Linear Algebra, Real Analysis

EXPERIENCE

Google Jun 2026 – Present

Mentee, Virtual Software Engineering Program (V-SWEP) *Remote*

- Selected for Google's competitive mentorship program for CS students; paired 1:1 with a Google software engineer for technical and career mentorship; returned for the Fall 2026 session

Unadat Mar 2026 – May 2026

Software Engineering Intern *New York, NY*

- Rebuilt the mobile landscape layout of the Chores feature (task tracking for business clients) to resize responsively with touch-specific targeting, fixing all broken mobile layouts with no desktop regressions
- Co-developed ticketing, notification, and scoring systems for business clients in PHP, JavaScript, and MySQL with a cross-functional team of 10 engineers over 7 sprints
- Found 6 bugs in layout, data handling, and alignment through cross-device and cross-browser testing

PROJECTS

Kalshi Market-Making & Adverse Selection Analysis | *CUNY Tech Prep, Data Science Track* Sep 2026 – Present

- Quantifying when market making on Kalshi is profitable in a team of 5, using a self-collected 24/7 record of the full order book across ~250 markets in 14 categories (~2.2 GB/day; 97.9% match to Kalshi's own quotes)
- Measuring adverse selection as effective minus realized spread, with markouts from 5 seconds to 5 minutes, net of fees and incentive payments, using point-in-time joins to prevent look-ahead bias
- Building a gradient-boosting model to predict toxic order flow and backtesting quoting strategies against naive and Avellaneda-Stoikov baselines in Python and DuckDB; presenting findings at LinkedIn NYC in Dec 2026

Severity-Adjusted Hospital Benchmarking (NY SPARCS) | *Independent Study* Sep 2026 – Present

- Building a PySpark and SQL pipeline to ingest and clean New York State's public SPARCS hospital discharge data (2M+ inpatient stays per year), scaling from one year to multiple years with automated data-quality checks
- Modeling expected length of stay and cost per admission from diagnosis group and severity of illness, setting a risk-adjusted baseline for every hospital
- Comparing each hospital's observed outcomes to expected, adjusting for low-volume hospitals, and presenting results in a dashboard and written report

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript, PHP, C/C++, HTML/CSS

Data & ML: pandas, NumPy, scikit-learn, PySpark, DuckDB, Matplotlib, Jupyter

Statistics: Regression, Classification, Hypothesis Testing, Statistical Modeling

Tools: Git, GitHub, MySQL, REST APIs, Excel